VIETNAM GENERAL CONFEDERATION OF LABOR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY



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FINAL REPORT

REQUIREMENTS ANALYSIS AND DESIGN

HO CHI MINH CITY, 2026

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Advised by
Dr. Ho Thi Linh

HO CHI MINH CITY, 2026

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Ho Chi Minh city, 22nd October 2026.

Author

(Signature and full name)

Ly Phuc Hai

Le Hoai An

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CHAPTER 1. INTRODUCTION

1.1 Background and objectives

Beauty and personal-care services in Vietnam are growing quickly, and the number of spas, salons, nail studios and cosmetic clinics keeps rising. Most small and medium-sized businesses still manage operations with notebooks, Excel files, Zalo/Facebook messages and the receptionist's memory. This causes **double-booked appointments**, customers arriving when no technician is free, hard-to-track working hours and commissions, lost supplies, and weak customer retention because there is no consistent **loyalty program**.

SpaCare is a spa/salon management system that digitizes the whole chain: service booking, staff management, payment and customer loyalty. The problem is also well suited to requirements analysis and design because it involves several actors, time and resource constraints (staff, rooms) and business rules for points, membership tiers and commissions.

The objectives are: (1) survey and describe the operations of a mid-sized spa/salon; (2) identify, specify and prioritize functional and non-functional requirements; (3) model the system with UML and an ERD; (4) design the database and the user interface; (5) build a UI prototype and a running Java program for the core functions.

1.2 System scope

The system **does**:

- Manage customers, services, combos and treatment packages.
- Book appointments (at the counter, by phone, online), detect scheduling conflicts and send reminders.
- Manage staff, shifts, attendance and commissions.
- Handle payment, invoicing, discounts and promotions.
- Run a loyalty program: points, membership tiers, point redemption and vouchers.

- Manage supplies/products at a basic level and provide statistical reports.

The system **does not**:

- Provide full financial accounting (general ledger, tax, detailed payroll); it only exports revenue and commission data.
- Handle medical records or clinical consultation of cosmetic clinics.
- Offer e-commerce with delivery, or integrate real payment gateways (simulated at prototype level only).

1.3 Submission structure

```
SpaCare_Final_Submission/
├── 01_Report/              report (Word and PDF)
├── 02_Slides/              presentation slides
├── 03_Demo/                demo only: one folder per function
│   ├── 01_Login/           UI/index.html   Java/ (run.sh)   GUIDE.md
│   ├── 02_Booking_Conflict/
│   ├── 03_Reschedule_Cancel/
│   ├── 04_CheckIn_Service/
│   └── 05_Payment_Loyalty/
├── 04_Database/            schema.sql, data.sql
├── 05_Presentation_Script/ timed script and demo script
├── 06_QA_Preparation/      prepared questions and answers
└── 07_Diagrams/            UML/ERD images
```

1.4 Coverage of the assignment requirements

The table maps every mandatory item of the assignment to the place where it is covered, so that the report can be checked against the requirements quickly.

Required item	**Where covered**
Cover page, inner cover, table of contents, lists of figures/tables, page numbering	Front matter
Project introduction: rationale, objectives, scope	Chapter 1
Survey and description of the current state	Chapter 2
Functional requirements (FR-01...), non-functional requirements, MoSCoW prioritization	Chapter 3
Use Case diagram, 7 use case specifications, ERD, 3 sequence/activity diagrams	Chapter 4
Database design (tables, keys, constraints, data dictionary)	Chapter 5
User interface design: wireframes and navigation flow	Chapter 6

Required item	Where covered
Prototype plan/results: UI prototype and Java demo, demo script	Chapter 7, Chapter 8
Conclusion: results, limitations, future work	Chapter 9
Task assignment table and references	Appendix A, References

Table 1.1: Coverage of the assignment requirements

CHAPTER 2. CURRENT-STATE SURVEY AND DESCRIPTION

2.1 Description of business operations

The surveyed organization is a mid-sized spa/salon with about 10–20 staff, 6–10 beds/chairs and 40–80 customer visits per day. It offers facial care, body massage, hair services (wash, cut, styling, coloring), nails and multi-session treatment packages, and also sells skin and hair care products.

Note for the group: this chapter should be cross-checked with interviews or observation of one or two real spas/salons and cited in the references.

2.2 Stakeholders

Stakeholder	Role and concerns
Customers	Convenient booking, being served on time, earning points and receiving offers
Receptionist	Receives bookings, checks customers in, takes payment; needs fast, low-error operations
Technician/Stylist	Knows own schedule, confirms completed services, tracks commission
Manager/Owner	Schedules shifts, monitors revenue, staff performance and stock, configures promotions
Cashier/Accountant	Reconciles invoices and payments, revenue reports

Stakeholder	Role and concerns
SMS/Zalo/Email provider (external)	Sends appointment reminders and promotions

Table 2.1: Stakeholders

2.3 Current business processes

- **Booking:** customers call, message on Zalo/Facebook or walk in. The receptionist records the booking in a notebook or Excel file and asks staff availability through a group chat.
- **Reminders:** done manually by phone or message, often forgotten.
- **Reception and service delivery:** the receptionist finds the name in the notebook, assigns a bed and a staff member, and staff tell the receptionist verbally when finished.
- **Payment:** the bill is added up with a calculator, written by hand or printed from simple POS software; discounts are applied from memory.
- **Loyalty:** paper cards or a notebook, easily lost and hard to verify.
- **Attendance, commission and reporting:** compiled by the manager at month end from handwritten notes and consolidated manually in Excel.

2.4 Problems and limitations to be addressed

No.	Problem	Consequence
1	Manual booking with no automatic conflict check	Double-booked staff/beds; customers wait or complain
2	No automatic reminders	High no-show rate
3	Scattered customer data, no service history	Hard to personalize or advise
4	Manual points and tiers	Errors, loss of trust, poor retention
5	Hand-calculated commissions and working hours	Slow; disputes with staff
6	No tracking of prepaid treatment packages	Customers forget remaining sessions; hard reconciliation
7	Supplies not deducted per service	Losses, unexpected stock-outs

No.	Problem	Consequence
8	Slow, non-real-time reports	Decisions made without data

Table 2.2: Problems of the current process

CHAPTER 3. REQUIREMENTS IDENTIFICATION AND SPECIFICATION

3.1 Functional requirements

ID	Group	Requirement
FR-01	Account	Log in/out and role-based authorization (Manager, Receptionist, Technician, Customer)
FR-02	Customer	Create, search and update customer profiles (name, phone, birthday, notes, allergies) and view service history
FR-03	Service	Manage service categories, services and combos (price, duration, required room/equipment)
FR-04	Appointment	Create an appointment: choose customer, services, staff (or "any"), date and time
FR-05	Appointment	Suggest free time slots and check conflicts for staff and rooms/beds
FR-06	Appointment	Reschedule or cancel appointments according to policy (at least 2 hours in advance) and record the reason
FR-07	Appointment	Send automatic reminders via SMS/Zalo/Email before the appointment
FR-08	Service delivery	Check in customers and update status (confirmed, checked-in, in progress, completed, cancelled, no-show)
FR-09	Staff	Manage staff profiles and skills (services each member may perform)
FR-10	Staff	Assign shifts and manage work schedules and days off
FR-11	Staff	Record attendance and calculate commissions per service/product
FR-12	Payment	Create invoices from appointments; pay by cash, card, transfer/QR; print or export invoices
FR-13	Loyalty	Automatically earn points based on invoice value
FR-14	Loyalty	Manage membership tiers (Silver, Gold, Platinum), automatic upgrade and tier benefits
FR-15	Loyalty	Redeem points for discounts/vouchers; manage promotions

ID	Group	Requirement
FR-16	Packages	Sell and track multi-session packages (remaining sessions, expiry date)
FR-17	Inventory	Manage supplies/products: receipts, issues, stock, low-stock alerts; deduct supplies per service
FR-18	Feedback	Customers rate services and staff after completion
FR-19	Reporting	Revenue, staff performance, no-show rate and loyal-customer reports by period

Table 3.1: Functional requirements

3.2 Non-functional requirements

ID	Type	Description
NFR-01	Performance	Search and booking operations respond in under 2 seconds with 50 concurrent users
NFR-02	Security	Hashed passwords, role-based access control (RBAC), account lockout after 5 failed logins
NFR-03	Data protection	HTTPS in transit; customers' personal data protected in line with Vietnam's Decree 13/2023/ND-CP
NFR-04	Availability	At least 99% uptime during opening hours; daily backups
NFR-05	Scalability	Support multiple branches; add services, tiers or reminder channels without changing core code
NFR-06	Usability	Vietnamese interface; a new receptionist can learn it in under 1 hour; works on desktop and tablet
NFR-07	Integrity	No double booking of a staff member or room in the same time slot; all point and payment transactions are logged
NFR-08	Maintainability	Layered architecture, commented code, unit tests for core business logic
NFR-09	Compatibility	Runs on Windows/macOS/Linux with JDK 17+; MySQL 8 or PostgreSQL database

Table 3.2: Non-functional requirements

3.3 Requirements prioritization (MoSCoW)

Priority	Requirements	Rationale
Must have	FR-01, 02, 03, 04, 05, 06, 08, 09, 10, 12, 13	Core operations; the system cannot run without them

Priority	Requirements	Rationale
Should have	FR-07, 11, 14, 15, 16, 19	High value and differentiating; can be handled manually in the short term
Could have	FR-17, 18	Improve experience and control; deferred to a later phase
Won't have (this release)	Separate mobile app, real payment gateway, full accounting	Out of scope; future directions

Table 3.3: MoSCoW prioritization

CHAPTER 4. REQUIREMENTS MODELING WITH UML

4.1 Actors

Actor	Description
Customer	Uses services; books online, views points and tier, leaves reviews
Receptionist	Takes bookings, checks customers in, handles payment, manages customer profiles
Technician/Stylist	Views personal schedule, updates service status
Manager	Manages services, staff, shifts, promotions, stock; views reports
Notification service (external)	Sends SMS/Zalo/Email reminders

Table 4.1: Actors

4.2 Overall Use Case model

ID	Use case	Primary actor	Related requirements
UC-01	Log in and authorize	All	FR-01
UC-02	Manage customers	Receptionist, Manager	FR-02
UC-03	Manage services/combos	Manager	FR-03
UC-04	Book appointment	Customer, Receptionist	FR-04, 05
UC-05	Reschedule/cancel appointment	Customer, Receptionist	FR-06

ID	Use case	Primary actor	Related requirements
UC-06	Check in and perform service	Receptionist, Technician	FR-08
UC-07	Pay and issue invoice	Receptionist	FR-12
UC-08	Earn and redeem points	Receptionist, Customer	FR-13, 14, 15
UC-09	Manage staff	Manager	FR-09
UC-10	Assign shifts and record attendance	Manager	FR-10, 11
UC-11	Manage promotions and membership tiers	Manager	FR-14, 15
UC-12	Manage treatment packages	Receptionist, Manager	FR-16
UC-13	Manage inventory	Manager	FR-17
UC-14	View reports	Manager	FR-19
UC-15	Review service	Customer	FR-18
UC-16	Send reminders	Notification service	FR-07
UC-17	Check scheduling conflicts	(System)	FR-05

Table 4.2: Use case list

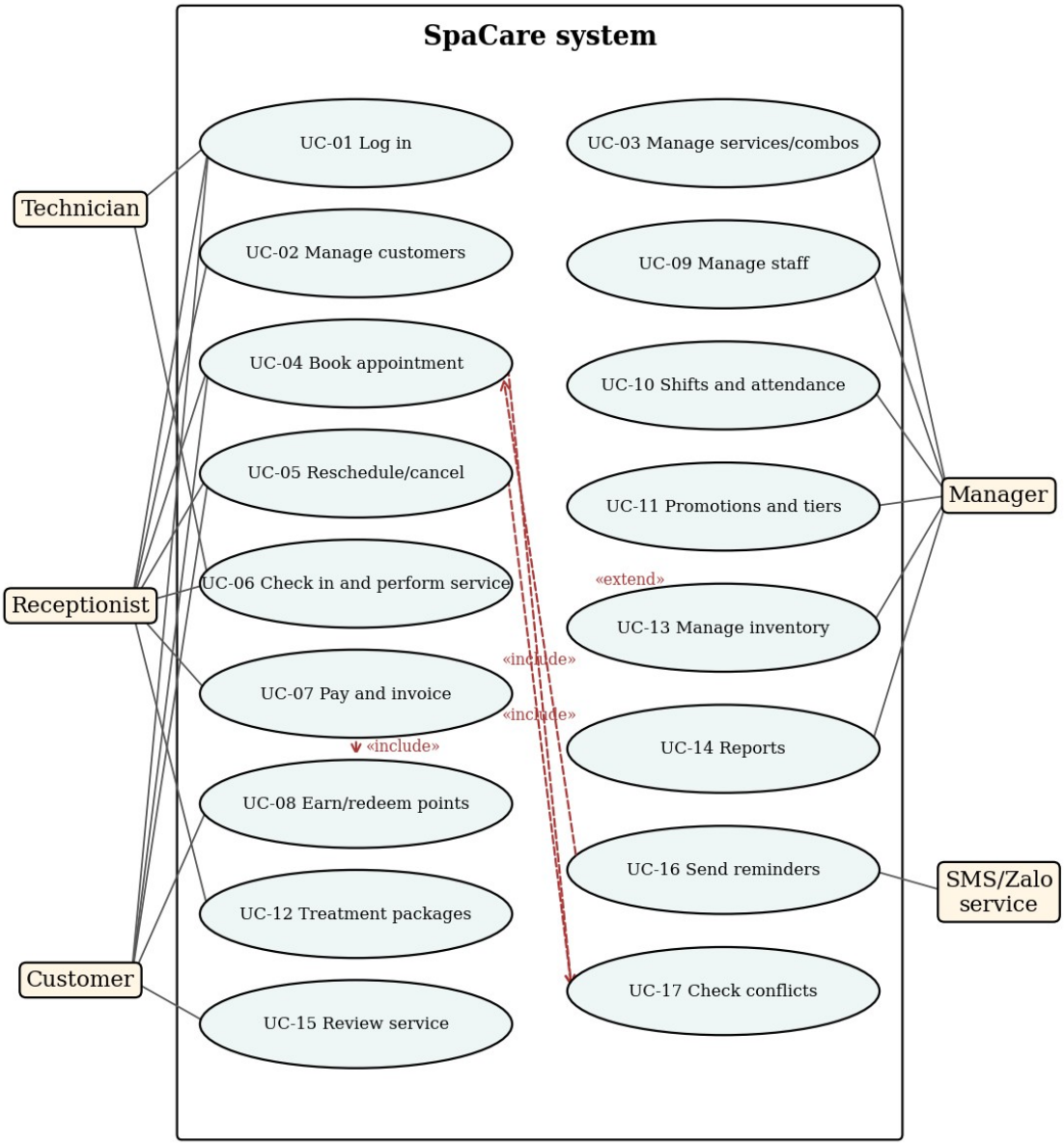


Figure 4.1: Overall Use Case diagram

4.3 Detailed Use Case specifications

Item	Content
Actors	Customer, Receptionist, Technician, Manager
Description	A user authenticates to access the functions allowed for their role
Preconditions	The account exists and is active
Postconditions	A session is created; the menu reflects the user's role

Item	Content
Main flow	1. The user opens the login screen. 2. Enters username and password. 3. The system verifies the credentials (hash comparison). 4. The system determines the role and opens the matching home page
Alternative flow	2a. The user selects "Forgot password": the system sends a reset link by email/SMS
Exception flow	3a. Wrong credentials: show an error and increase the failure counter. 3b. 5 failures: lock the account for 15 minutes. 3c. Disabled account: deny access

Table 4.3: Specification of UC-01 (Log in and authorize)

Item	Content
Actors	Receptionist, Customer (online booking)
Description	Create an appointment for one or more services with a suitable staff member and time slot
Preconditions	The actor is logged in; at least one service is active; staff have assigned shifts
Postconditions	The appointment is saved as CONFIRMED; staff and room are reserved; a reminder is scheduled
Main flow	1. Select or create the customer. 2. Select services/combo. 3. Select a staff member or "any". 4. Select the date. 5. The system sums the duration and shows free slots (UC-17). 6. Select a slot. 7. Add a note (optional) and confirm. 8. The system re-checks conflicts, saves the appointment and sends a confirmation
Alternative flow	3a. "Any": the system assigns a qualified, free staff member. 2a. The customer uses a package: select a session from a valid package
Exception flow	5a. No free slot: suggest other dates/staff. 8a. A conflict arises because someone just booked: show an error and return to step 5. 8b. The customer already has an overlapping appointment: warn and ask for confirmation

Table 4.4: Specification of UC-04 (Book appointment)

Item	Content
Actors	Customer, Receptionist
Description	Change the time of, or cancel, an existing appointment
Preconditions	The appointment status is PENDING or CONFIRMED

Item	Content
Postconditions	The appointment is updated (reschedule) or set to CANCELLED; the old slot is released
Main flow	1. Find the appointment by phone number or code. 2. Choose "Reschedule" or "Cancel". 3. The system checks the policy (at least 2 hours before the start time). 4. Reschedule: choose a new slot (UC-17). Cancel: enter a reason. 5. The system saves the change and notifies the customer
Alternative flow	3a. The receptionist/manager may override the policy with a reason
Exception flow	3b. Past the deadline: reject and advise contacting the venue. 4a. The new slot conflicts: return to selection

Table 4.5: Specification of UC-05 (Reschedule/cancel appointment)

Item	Content
Actors	Receptionist, Technician
Description	Confirm the customer's arrival and track service status
Preconditions	A CONFIRMED appointment exists for today
Postconditions	Status moves CHECKED_IN, IN_PROGRESS, then COMPLETED; actual times are recorded; supplies are deducted
Main flow	1. The receptionist finds the appointment and clicks Check-in. 2. The system notifies the technician. 3. The technician clicks "Start". 4. When finished, clicks "Complete". 5. The system records it and deducts supplies by usage norms
Alternative flow	1a. Walk-in customer: quickly create an appointment, then check in
Exception flow	1b. Customer more than 15 minutes late: keep or reschedule the slot. 1c. Customer does not show up: status becomes NO_SHOW after the slot ends

Table 4.6: Specification of UC-06 (Check in and perform service)

Item	Content
Actors	Receptionist
Description	Create an invoice for completed services, apply benefits and collect payment
Preconditions	At least one COMPLETED service is unpaid
Postconditions	A PAID invoice is stored; points are earned (UC-08); commissions are recorded

Item	Content
Main flow	1. Select the appointment to pay. 2. The system creates a draft invoice. 3. Add retail products if any. 4. Apply tier discount, voucher and point redemption. 5. Choose the payment method and enter the amount. 6. The system confirms payment and closes the invoice. 7. Print or send the e-invoice. 8. The system adds points and updates the tier
Alternative flow	5a. Split payment across methods. 5b. Deduct a session from a treatment package instead of paying
Exception flow	4a. Voucher expired or used: show an error. 5c. Underpayment: keep the invoice as PARTIAL. 6a. Transfer cannot be confirmed: allow retry

Table 4.7: Specification of UC-07 (Pay and issue invoice)

Item	Content
Actors	Receptionist, Customer, System
Description	Add points by invoice value, upgrade tiers, redeem points for discounts
Preconditions	The customer is a member; the invoice is paid (for earning)
Postconditions	Point balance and tier are updated; the point transaction is logged
Main flow (earn)	1. The system takes the invoice value after discounts. 2. Points = value / 10,000 VND, multiplied by the tier factor (Silver 1.0; Gold 1.2; Platinum 1.5). 3. Points are added to the account. 4. The system sums spending over the last 12 months and upgrades the tier at the thresholds (Gold from 10 million VND, Platinum from 30 million VND)
Main flow (redeem)	1. The receptionist/customer chooses points to redeem at payment. 2. The system converts them (100 points = 10,000 VND), up to 50% of the invoice. 3. Points are deducted and logged
Exception flow	Not enough points: reject. Invoice refunded or voided: earned points are reversed. Points older than 12 months expire automatically

Table 4.8: Specification of UC-08 (Earn and redeem points)

Item	Content
Actors	Manager
Description	Schedule staff and record check-in/out times
Preconditions	Staff are active; shifts are defined

Item	Content
Postconditions	The schedule is saved; attendance data feeds payroll and commission
Main flow	1. The manager selects the week and branch. 2. Assigns staff to shifts per day. 3. The system verifies no overlapping shifts and enough days off. 4. Save and publish. 5. Staff clock in/out daily
Alternative flow	Shift swap between two staff: manager approval
Exception flow	Assigning a shift to staff on leave: reject. Deleting a shift that has appointments: reschedule the appointments first

Table 4.9: Specification of UC-10 (Assign shifts and record attendance)

The tier thresholds, point multipliers and redemption rules above are the group's business assumptions and can be changed in the membership tier configuration.

4.4 Data model (ERD)

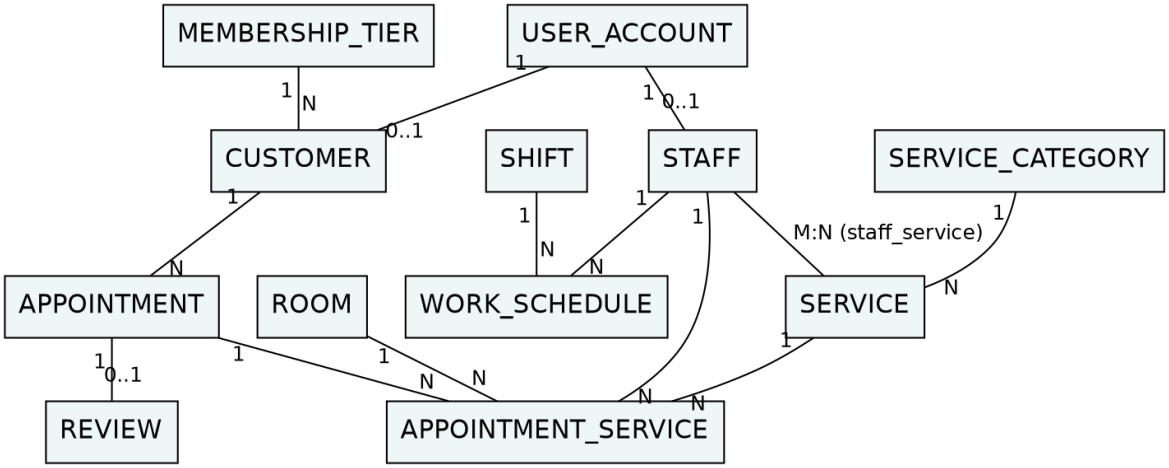


Figure 4.2: ERD part 1 – customers, staff, services and appointments

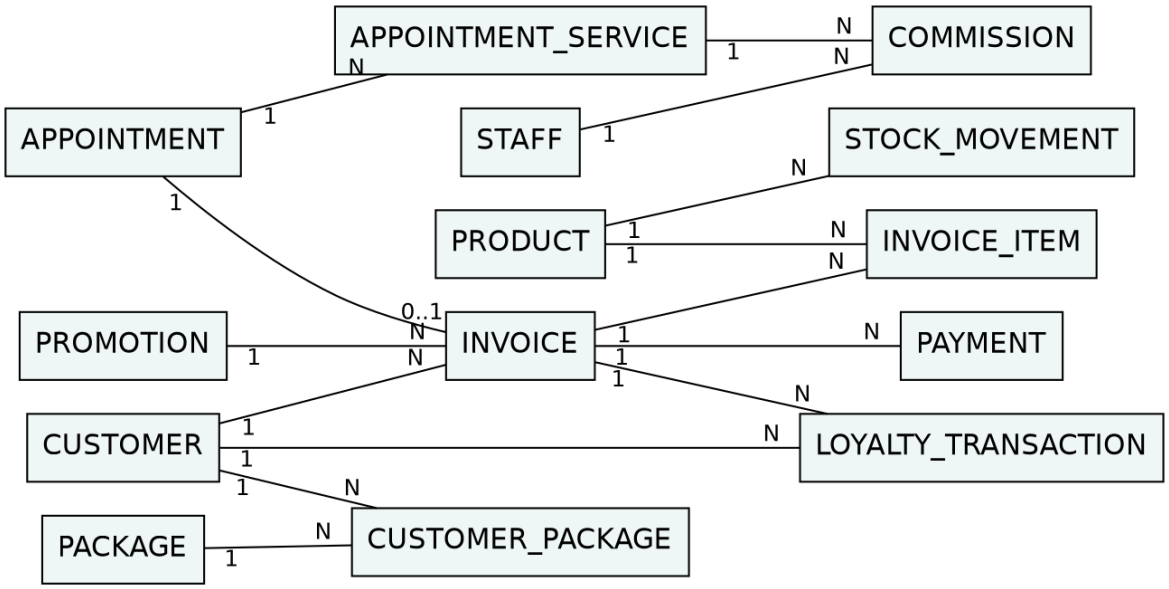


Figure 4.3: ERD part 2 – billing, loyalty, packages, inventory and commissions

4.5 Sequence and activity diagrams

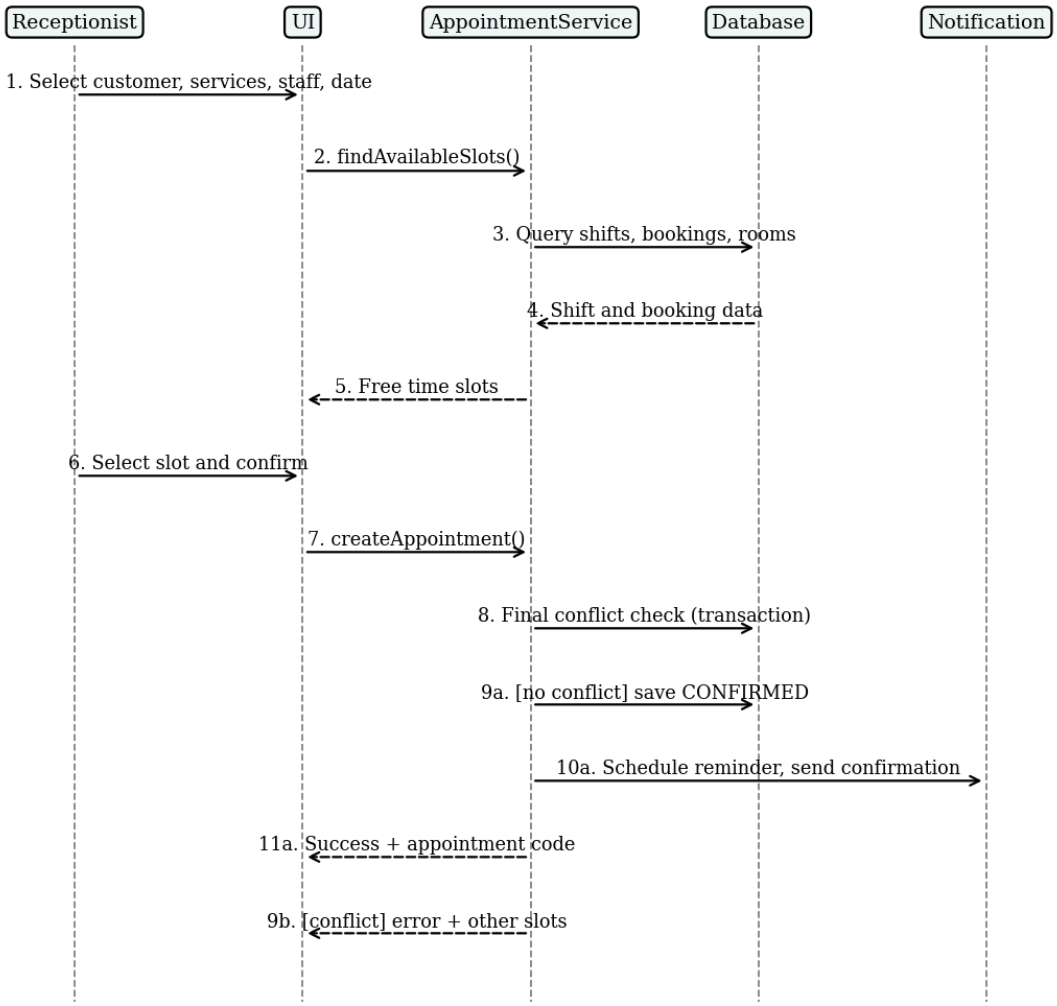


Figure 4.4: Sequence diagram – Book appointment (UC-04)

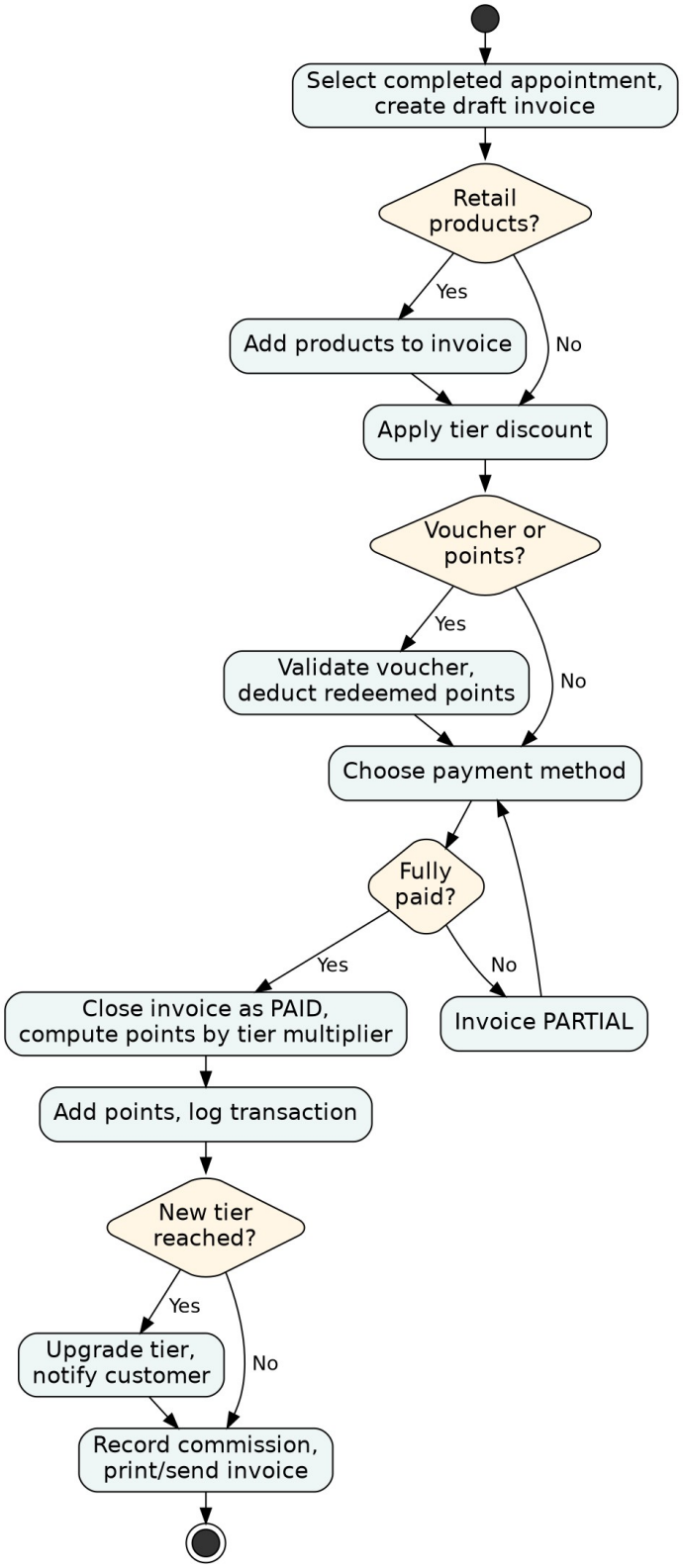


Figure 4.5: Activity diagram – Payment and loyalty points (UC-07, UC-08)

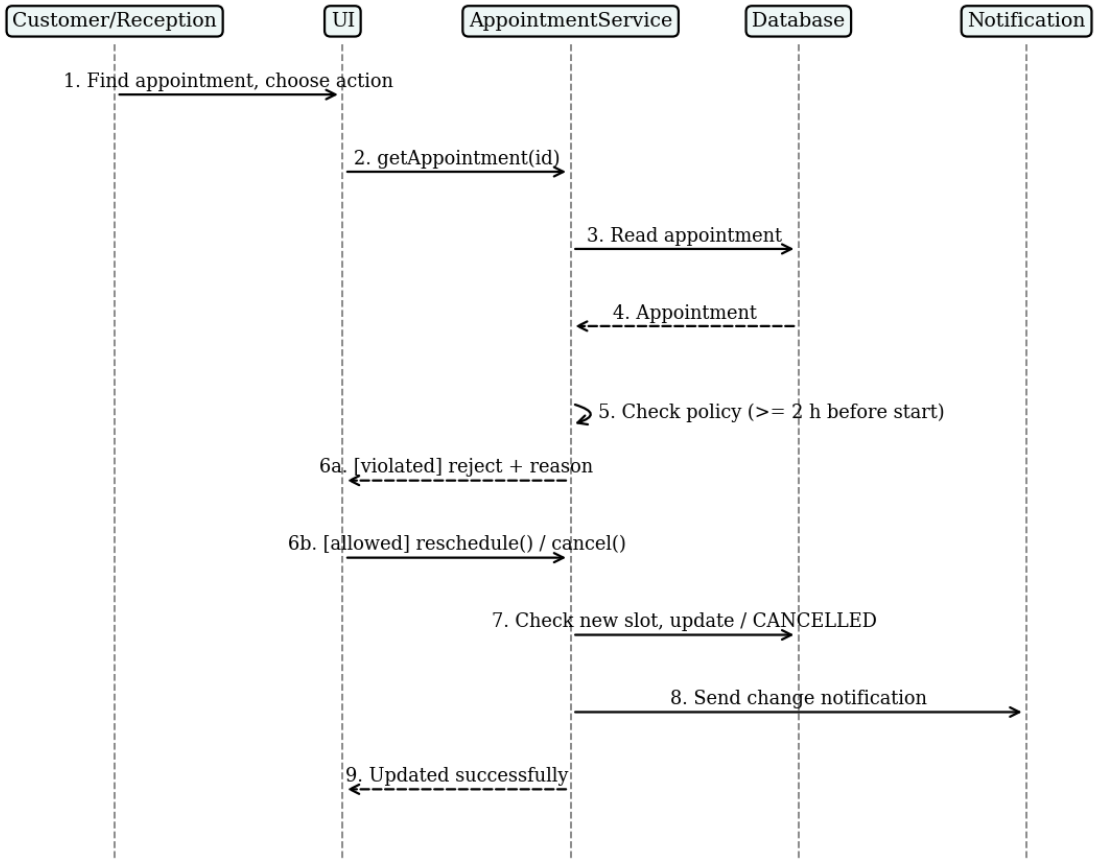


Figure 4.6: Sequence diagram – Reschedule/cancel appointment (UC-05)

CHAPTER 5. DATABASE DESIGN

5.1 Design principles

The database is relational (MySQL 8 or PostgreSQL) and normalized to third normal form (3NF). Primary keys are auto-increment integers; table and column names are English snake_case, consistent with the Java source code. Notation: **PK** primary key, **FK** foreign key, **UQ** unique, **NN** not null.

5.2 Relational schema

Table	Main columns	Constraints
user_account	id PK, username UQ NN, password_hash NN, role NN, is_active, failed_attempts, locked_until	role limited to ADMIN, MANAGER, RECEPTIONIST, TECHNICIAN, CUSTOMER

Table	Main columns	Constraints
membership_tier	id PK, name UQ NN (SILVER, GOLD, PLATINUM), min_spending, point_multiplier, discount_percent	min_spending ≥ 0; point_multiplier > 0
customer	id PK, user_account_id FK UQ, tier_id FK NN, full_name NN, phone UQ NN, email, birth_date, allergy_note, point_balance, spending_12m, created_at	point_balance ≥ 0
staff	id PK, user_account_id FK UQ, full_name NN, phone, position, hire_date, base_commission_rate, status	status in ACTIVE, ON_LEAVE, RESIGNED
branch	id PK, name NN, address, phone	Supports multiple branches (NFR-05)

Table 5.1: Accounts, customers, staff

Table	Main columns	Constraints
service_category	id PK, name UQ NN	–
service	id PK, category_id FK NN, name NN, price NN, duration_min NN, room_type, is_active	price ≥ 0; duration_min > 0
staff_service	staff_id FK, service_id FK	Composite PK (staff_id, service_id)
room	id PK, branch_id FK NN, name NN, room_type, is_active	UQ (branch_id, name)
appointment	id PK, customer_id FK NN, branch_id FK NN, start_time NN, end_time NN, status NN, channel, note, cancel_reason, created_by FK	status in PENDING, CONFIRMED, CHECKED_IN, IN_PROGRESS, COMPLETED, CANCELLED, NO_SHOW; end_time > start_time

Table	Main columns	Constraints
appointment_service	id PK, appointment_id FK NN, service_id FK NN, staff_id FK NN, room_id FK, start_time NN, end_time NN, price_at_booking NN, customer_package_id FK	No overlapping interval for the same staff_id or room_id

Table 5.2: Services and appointments

Table	Main columns	Constraints
promotion	id PK, code UQ NN, type, value NN, valid_from, valid_to, usage_limit, min_invoice	value > 0; valid_to ≥ valid_from
invoice	id PK, appointment_id FK UQ, customer_id FK NN, promotion_id FK, subtotal NN, tier_discount, promo_discount, point_discount, total NN, status NN, created_at, cashier_id FK	status in DRAFT, PARTIAL, PAID, VOID; total ≥ 0
invoice_item	id PK, invoice_id FK NN, item_type, service_id FK, product_id FK, quantity NN, unit_price NN	quantity > 0; exactly one of service_id/product_id is set
payment	id PK, invoice_id FK NN, method NN, amount NN, paid_at, reference_code	method in CASH, CARD, TRANSFER, QR; amount > 0
loyalty_transaction	id PK, customer_id FK NN, invoice_id FK, type NN, points NN, balance_after NN, created_at, expires_at	type in EARN, REDEEM, EXPIRE, ADJUST; balance_after ≥ 0
package	id PK, name NN, service_id FK NN, total_sessions NN, price NN, valid_days	total_sessions > 0
customer_package	id PK, customer_id FK NN, package_id FK NN, remaining_sessions NN, purchased_at, expires_at	0 ≤ remaining_sessions ≤ total_sessions

Table 5.3: Invoices, payments and loyalty

Table	Main columns	Constraints
shift	id PK, name NN, start_time NN, end_time NN	end_time > start_time

Table	Main columns	Constraints
work_schedule	id PK, staff_id FK NN, shift_id FK NN, work_date NN, check_in, check_out, status	UQ (staff_id, work_date, shift_id)
commission	id PK, staff_id FK NN, appointment_service_id FK, invoice_item_id FK, amount NN, period	amount ≥ 0
product	id PK, name NN, sku UQ, unit, sale_price, stock_qty NN, reorder_level	stock_qty ≥ 0
service_product	service_id FK, product_id FK, qty_per_use NN	Composite PK; usage norm per service
stock_movement	id PK, product_id FK NN, type, quantity NN, reason, created_at	type in IN, OUT; quantity > 0
review	id PK, appointment_id FK UQ, rating NN, comment, created_at	rating between 1 and 5

Table 5.4: Shifts, commissions, inventory, reviews

5.3 Key integrity rules

- **Conflict prevention:** within one transaction, before inserting an appointment_service row, verify that no non-cancelled row for the same staff_id (or room_id) overlaps the interval [start_time, end_time). PostgreSQL can add an EXCLUDE constraint on time ranges; with MySQL use row locking (SELECT ... FOR UPDATE).
- **Points integrity:** customer.point_balance is a derived total and is always updated in the same transaction that writes the loyalty_transaction row.
- **Soft delete:** customers, staff and services with history are never physically deleted, only deactivated.
- **Passwords:** only hashes are stored, never plain text.

5.4 Data dictionary (excerpt)

Attribute	Table	Type	Meaning
status	appointment	VARCHAR(15)	Appointment lifecycle state
duration_min	service	INT	Service duration in minutes, used to compute end_time

Attribute	Table	Type	Meaning
point_multiplier	membership_tier	DECIMAL(3,2)	Points multiplier by tier
spending_12m	customer	BIGINT	Spending over the last 12 months, used for tier evaluation
price_at_booking	appointment_service	BIGINT	Price at booking time, unaffected by later price changes
balance_after	loyalty_transaction	INT	Point balance after the transaction, for reconciliation

Table 5.5: Data dictionary (excerpt)

The demo program uses a simplified subset of this schema (resources/schema.sql): the appointment table keeps start_time/end_time directly and branch_id is omitted.

CHAPTER 6. USER INTERFACE DESIGN

6.1 Design principles

The interface is in Vietnamese and keeps the number of steps low for receptionists. Calm colors (teal, beige) match a spa brand, appointment states are distinguished by both color and text label, and layouts work on desktop and tablet. The wireframes below are sketches; the colored mockups are built in Figma (see Chapter 7).

6.2 Navigation flow

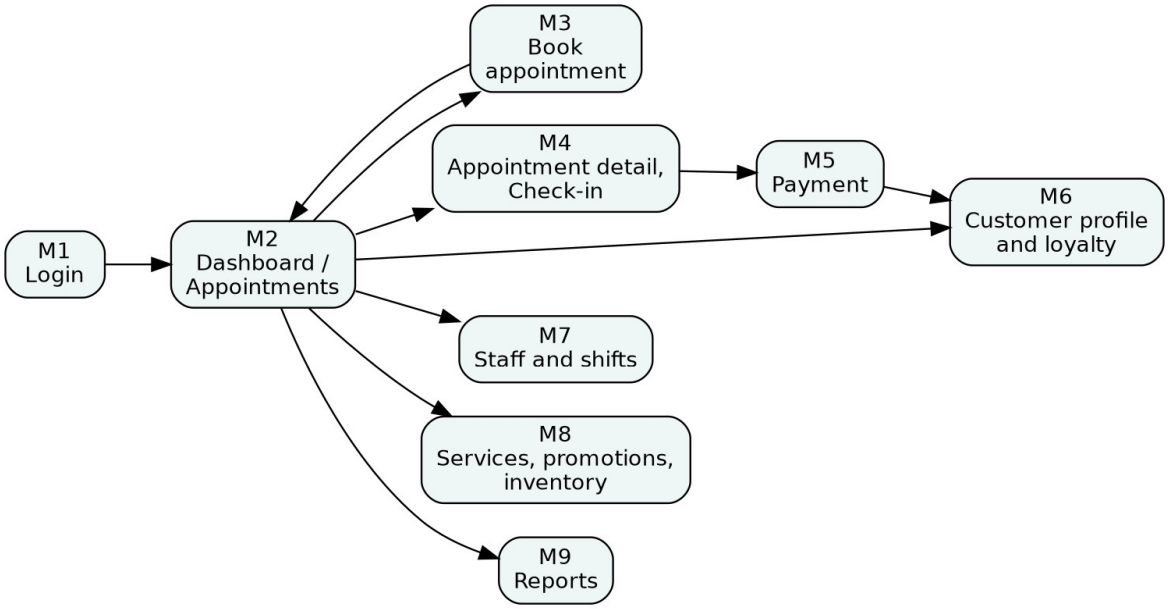


Figure 6.1: Navigation flow between the main screens

6.3 Wireframes of the main screens

```
+--------------------------------------+
|            [SpaCare logo]            |
|  Username: [_____________________]   |
|  Password: [_____________________]   |
|  [ ] Remember me     Forgot password?|
|              [   LOG IN   ]          |
+--------------------------------------+
```

Figure 6.2: M1 – Login

```
+-------------------------------------------------------------+
| SpaCare | Appointments | Customers | Staff | Reports | [User]  |
+-------------------------------------------------------------+
| < Today 12/11/2026 >  Branch: [D1 v]   [+ New appointment]   |
+------+---------------+---------------+----------------+----+
| Time | Lan (Spa)     | Minh (Stylist) | Ha (Nail)      |    |
| 09:00| [Massage-An]  |                | [Nail-Chi]     |    |
| 10:00| [Massage-An]  | [Cut+Wash-Binh]|                |    |
| 11:00|               | [Color-Hoa]    | [Gel-Mai]      |    |
+------+---------------+---------------+----------------+----+
Legend: Confirmed | Checked-in | In progress | Completed
```

Figure 6.3: M2 – Daily appointments (receptionist's main screen)

```
+-----------------------------------------------------+
| BOOK APPOINTMENT                                    |
| Customer: [Search phone/name__] [+ New]   Tier: Gold |
| Services: [Select service/combo v] [+ add]          |
|           Relaxing hair wash  60m  150,000d   [x]   |
| Staff:    ( ) Any  ( ) [Select staff v]             |
| Date: [12/11/2026]   Free slots: [09:00][10:30]...  |
| Note: [_____________________________]               |
|                      [ Cancel ]   [ CONFIRM BOOKING ]|
+-----------------------------------------------------+
```

Figure 6.4: M3 – Book appointment

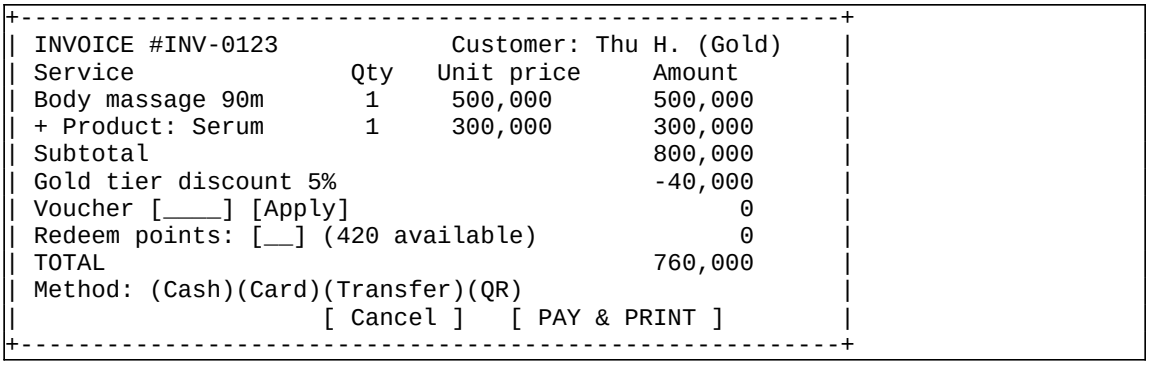


Figure 6.5: M5 – Payment

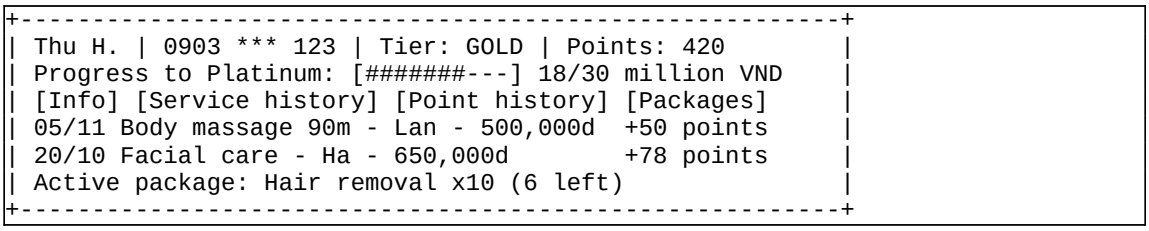


Figure 6.6: M6 – Customer profile and loyalty

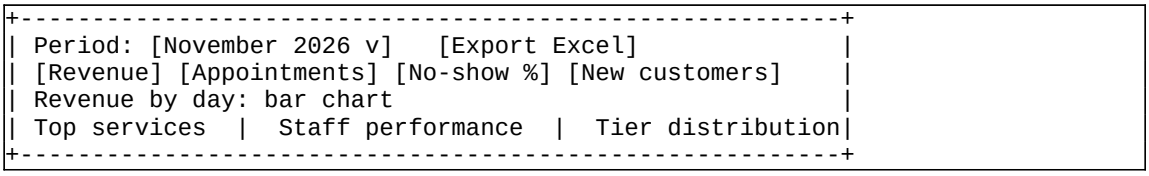


Figure 6.7: M9 – Reports

CHAPTER 7. PROTOTYPE PLAN AND DEMO

7.1 Technologies

Component	Choice
UI prototype	Single-file HTML prototypes (clickable), one per function; Figma optional
Demo program language	Java 17
Interface	Console application (alternatives: JavaFX/Swing, Spring Boot + Thymeleaf)
Database	H2 in-memory for the demo, MySQL 8 for production, accessed via JDBC
Project tooling	Maven, Git (GitHub/GitLab)

Component	Choice
Testing	JUnit 5 for core logic (conflict checking, points calculation, tier upgrade) – planned

Table 7.1: Technologies used

The program follows a layered architecture: presentation (Main) → service (AppointmentService, BillingService) → data access (JDBC, class Db) → database. Separating the service layer allows independent testing and easy replacement of the interface.

7.2 Demo scope

The demonstration has two mandatory parts, and both are organized as five demos, one per main function. Every demo has its own folder in 03_Demo with a separate UI prototype (UI/index.html), a separate Java program (Java/) and a GUIDE.md.

Demo	Function (folder)	Use cases	UI prototype / Java program
1	Login and roles (01_Login)	UC-01	Role-based welcome, lockout after 5 failures / LoginDemo
2	Booking and conflict (02_Booking_Conflict)	UC-04, UC-17	Booking form with busy slots / BookingDemo
3	Reschedule and cancel (03_Reschedule_Cancel)	UC-05	Reschedule and cancel panel / RescheduleDemo
4	Check-in and service (04_CheckIn_Service)	UC-06	Status flow buttons / ServiceDemo
5	Payment and loyalty (05_Payment_Loyalty)	UC-07, UC-08	Invoice, points, tier upgrade / PaymentDemo

Table 7.2: Demo folders and the use cases they cover

UI/prototype part. The prototypes reflect the screens and business flows designed in Chapter 6 (M1 Login, M2 Appointments, M3 Book, M4 Check-in, M5 Payment, M6 Loyalty).

Java part. Real Java code runs live on the presenter's machine and implements the main use cases UC-01, 04, 05, 06, 07 and 08, which is 6 of the 7 specified use

cases (86%) and above the required 3–5 functions. UC-10 (shifts and attendance) is specified and its sample shift data is loaded, but its screens are left to future work.

7.3 Sample data

Data	Sample content
Accounts	admin, letan01, ktv_lan, ql_hoa (password 123456)
Services	Body massage 90m (500,000 VND), Relaxing hair wash 60m (150,000 VND), Basic facial 60m (400,000 VND), Women's haircut 45m (200,000 VND), Gel polish 60m (250,000 VND)
Staff	Lan (spa), Ha (nail/skin), Minh (stylist, morning shift only); other staff work 08:00–20:00
Customers	Thu H. (Gold, 420 points), Mai (Silver, 80 points, 9.5 million VND spent, close to Gold), Linh (Platinum)
Rooms	Room 1, Room 2, Nail area, Hair area

Table 7.3: Sample data

7.4 Demo script (12–15 minutes)

Step	Content	Presenter	Time
1	Problem, scope and requirements (slides)	Ly Phuc Hai	2 min
2	Use cases, ERD and database design (slides)	Le Hoai An	2 min
3	Demo 1 – Login and roles: UI then Java	Ly Phuc Hai	1.5 min
4	Demo 2 – Booking: UI then Java, deliberate double booking shows the conflict error	Ly Phuc Hai	2.5 min
5	Demo 3 – Reschedule and cancel: UI then Java	Ly Phuc Hai	1.5 min
6	Demo 4 – Check-in and service: UI then Java	Le Hoai An	1.5 min
7	Demo 5 – Payment and loyalty: UI then Java; Mai crosses the threshold and is upgraded to Gold	Le Hoai An	2.5 min
8	Conclusion, limitations, future work (slides)	Ly Phuc Hai	1 min

Table 7.4: Demo script

Sample data for each demo is listed in Table 7.3 and in the GUIDE.md of the demo folder; every demo starts from the same data so the results can be repeated. Total about 14.5 minutes.

7.5 Implementation plan

Week	Work	Output
1	Survey, scope	Chapters 1–2
2	Requirements analysis	Chapter 3
3–4	UML and ERD modeling	Chapter 4
5	Database design, schema and sample data	Chapter 5, SQL files
6	UI design, Figma	Chapter 6, Figma link
7–8	Java implementation of 5 functions, testing	Source code
9	Finalize report, rehearse presentation and demo	Report, slides

Table 7.5: Implementation plan

7.6 Results and submission links

- UI prototype: 03_Demo/0X_*/UI/index.html (open in any browser); online version: https://claude.ai/artifact/Hhosp81pQ4vsk1JUsfkBcx; Figma link (optional): [paste link here]
- Java source: 03_Demo/0X_*/Java in the submitted zip, or repository link: [paste GitHub/GitLab link here]
- Demo accounts and run instructions: see Chapter 8 and the README files.

CHAPTER 8. INSTALLATION AND RUN GUIDE

8.1 Installation

Install JDK 17 or later and Maven 3.8 or later. No database installation is needed: the demo uses H2 in memory and creates the schema and sample data on every start.

8.2 Running the demos

Each demo is started from its own folder in 03_Demo. The UI part needs no installation: open UI/index.html in a browser. The Java part is started with one command:

```
cd 03_Demo/02_Booking_Conflict/Java
./run.sh               (Windows: run.bat)
```

Each Java program starts a fresh in-memory database and uses tomorrow's date, so the demos can be run in any order. The Windows .bat file switches the console to UTF-8; in an IDE run the main class with -Dfile.encoding=UTF-8.

Demo	Folder / class	Use cases	What it proves
1	01_Login / LoginDemo	UC-01	Roles, wrong passwords rejected, lockout after 5 failures
2	02_Booking_Conflict / BookingDemo	UC-04, UC-17	Free slots, double booking refused, adjacent booking allowed, skill and shift checks
3	03_Reschedule_Cancel / RescheduleDemo	UC-05	Reschedule honors conflicts, cancel frees the slot, status rules
4	04_CheckIn_Service / ServiceDemo	UC-06	Status machine CONFIRMED, CHECKED_IN, IN_PROGRESS, COMPLETED
5	05_Payment_Loyalty / PaymentDemo	UC-07, UC-08	Tier discount, 50% redemption limit, points earned, Mai upgraded to Gold, invalid payments refused

Table 8.1: Demo folders and Java classes

Lines marked [BỊ TỪ CHỐI] in the output are intended: they show business rules rejecting invalid actions.

8.3 Using MySQL instead of H2

Create a database named spacare, add the mysql-connector-j dependency to pom.xml and start with -Ddb.url=jdbc:mysql://localhost:3306/spacare -Ddb.user=root -Ddb.pass=....

8.4 Common problems

Problem	Fix
Vietnamese text is garbled	Run chcp 65001 (Windows) or use -Dfile.encoding=UTF-8
Unsupported class file version	Use JDK 17 or later

Problem	Fix
mvn not found	Install Maven and add it to PATH, or run spacare.Main from an IDE
No free slots appear	Slots are only offered in the future and inside the staff shift; use tomorrow's date
"Conflict" error when booking	Expected when the staff member or all rooms are busy; choose another slot

Table 8.2: Common problems and fixes

CHAPTER 9. CONCLUSION

9.1 Achieved results

The report surveys the current state of a mid-sized spa/salon; defines 19 functional and 9 non-functional requirements prioritized with MoSCoW; models the system with 17 use cases (7 specified in detail), an ERD and three sequence/activity diagrams; designs a database of more than 20 tables with integrity constraints; designs six main screens with a navigation flow; and provides a Figma prototype and a Java program demonstrating the core functions.

9.2 Limitations

- Business rules (tier thresholds, point multipliers, cancellation policy) are the group's assumptions and have not been validated with many real venues.
- The Java program implements only part of the functions; reminders and electronic payment are simulated, and the interface is a console application.
- No load testing has been performed and there is no customer-facing mobile app yet.

9.3 Future directions

- Mobile/Web booking app and a Zalo Mini App for customers.
- Integration with payment gateways (VNPay, MoMo) and legally compliant e-invoicing.
- Service recommendations and no-show prediction using data analysis.
- Multi-branch chains with unified permissions and reports.

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[5] Object Management Group, Unified Modeling Language Specification, Version 2.5.1, 2017.

[6] Government of Vietnam, Decree 13/2023/ND-CP on personal data protection, 2023.

[7] Oracle, Java SE 17 Documentation; H2 Database Engine documentation.

[8] [Add: surveyed venues, reference spa-management software or websites, design templates, images and libraries used in the project.]

APPENDIX A. TASK ASSIGNMENT TABLE

No.	Full name	Student ID	Tasks	Contribution	Signature
1	Ly Phuc Hai	524H0013	Chapters 1–3; use case specs; Java: booking, conflict check, login	50%	
2	Le Hoai An	524H0077	Chapters 4–6; ERD and database; Figma prototype; Java: payment, loyalty points	50%	
			Total	100%	

Table A.1: Task assignment table (edit tasks and percentages to match the actual work)